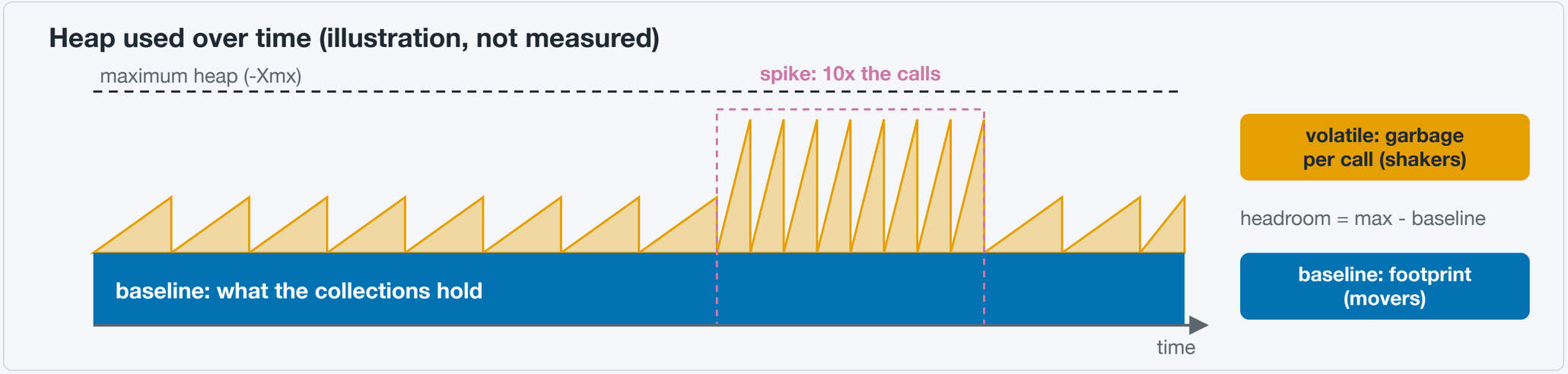
Two heaps in one

A heap has a baseline that stays and a volatile part that churns.

- Baseline: the live objects your collections hold between calls.
- Volatile: short-lived garbage that each call allocates and drops.
- The baseline sets the floor; the volatile part sets how often GC runs.
- Both have to fit under `-Xmx` at the busiest moment.

Baseline, churn and a spike



- Headroom is what is left above the baseline for garbage.
- A spike multiplies the garbage rate; the baseline stays where it was.

Why the volatile part is the risk

Stream garbage grows with calls per second, not with the number of elements.

- One JDK `anyMatch` call leaves 176 B, wherever the match is.
- At 10 M calls/s that is 1.76 GB/s of young-generation garbage.
- A 10 s spike at that rate allocates about 17.6 GB.
- The same question as an Eclipse Collections loop allocates 0 B.

References

- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)
- Factors Affecting Garbage Collection Performance (Java SE 25)
- The `java` command: `-Xmx` and GC options (Java SE 25)